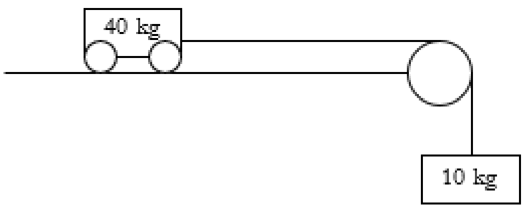
KINEMATICS PROBLEMS INVOLVING PULLEYS **03/11/25**

Draw force diagrams for these

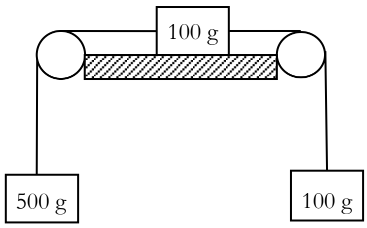
1. A truck of mass 40 kg can run smoothly on horizontal rails. A light, inextensible rope is attached to the front of the truck, and this runs parallel to the rails until it passes over a light, smoothly running pulley: the rest of the rope hangs down a vertical shaft, and carries a 10 kg load attached at the other end.



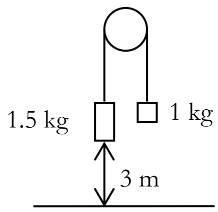
   (a) Find the tension in the rope and the acceleration of the truck and the load.
   (b) What hanging load would be necessary to give the truck an acceleration of 6 m s^{-2}?
   (c) What would be the acceleration if the truck were filled with 150 kg of sand?
   (d) With what force would the truck have to be pulled in order to raise the load with an acceleration of 2 m s^{-2}? What would be the tension in the rope then?
   (e) What would be the tension and the acceleration of the truck if its motion were hindered by a friction resistance of 25 N?

2. Masses of 5 kg and 3 kg are connected by a thread which runs over a smooth peg. They are released from rest when the 5 kg mass is 1·25 m above the floor. Assuming that the 3 kg mass does not first hit the pulley, how long does it take the 5 kg mass to reach the floor? (Take g = 10 m s^{-2}.)

3. Three weights, of masses 500 g, 100 g and 100 g, are connected by light inextensible strings running over smooth pulleys as shown in the diagram. The 100 g mass lies on a horizontal table and the other two masses hang freely. When the objects are released from rest and start to move, there is a constant resistance force of 2.8 N acting on the mass lying on the table. Find the acceleration of the objects and hence find the tension in both strings.



4. The following system is released from rest. Find the acceleration of the parcels and decide which will hit the ground first and after what time. With what speed does the parcel hit the ground? The pulley is smooth and freely turning; the string is light and the air resistance negligible.



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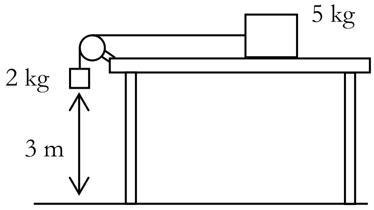
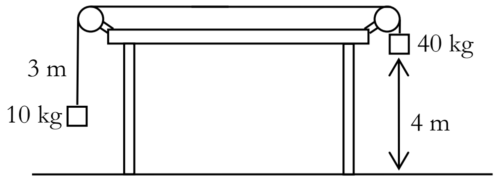
5. Find the time taken for the suspended parcel to hit the ground. Assume that the surface is smooth and horizontal.



6. In the following diagram, the left-hand parcel starts 1 m above the ground, and hits the pulley when it is 4 m above the ground. The string snaps and the pulley (which is of negligible mass) goes flying.



(a) Draw a velocity-time graph for each parcel up to the point where the string snaps.
(b) After the string breaks, both parcels are subject to an acceleration due to gravity. Continue your velocity time graphs until the parcels hit the ground.
(c) Find the time taken for each parcel to hit the ground and the speed at which it is travelling at impact.

7. Two empty buckets, each of mass 2 kg, have their handles connected by a rope which passes over a wheel free to turn smoothly about a horizontal axis. How much water must be poured into one of the buckets in order that they may move with an acceleration of $\frac{1}{5}g$? If twice this quantity of water is now poured into the other, with what acceleration will they move?

8. The diagram shows a man in a lift cage which is being raised by a counterweight. The lift cage and the counterweight are connected by a light inextensible cable which passes over a smooth pulley.

The man weighs 72 kg, the lift cage 48 kg and the counterweight 136 kg.
(a) Find the downward acceleration of the counterweight.
(b) What is the upward force exerted on the man by the lift cage?

9. Two unequal masses connected by a thread are slung over a smooth peg, and released so that each mass moves in a vertical line. Argue whether the tension in the thread is greater than, equal to, or less than the average weight of the two masses.